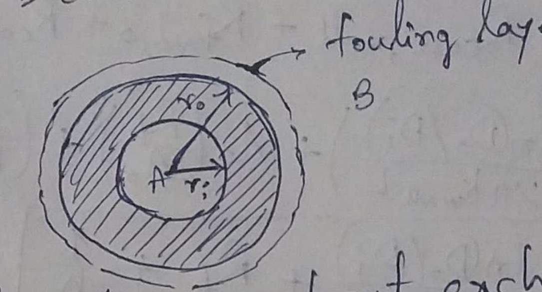
Part A :- Derivation of Overall Heat Transfer Coefficient

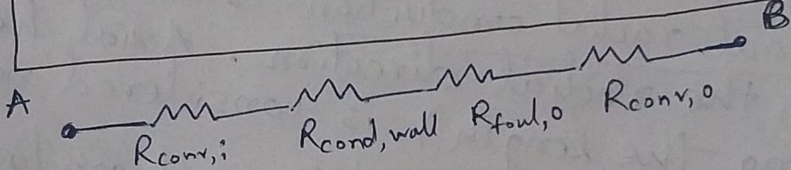


$D_o = 2r_o$; $D_i = 2r_i$

In a double pipe heat exchanger, heat flows radially from the hot fluid (inside), through the wall, to the cold fluid (outside).

We have to take the sum of thermal resistances (in series).

$$\boxed{R_{total} = R_{convection,i} + R_{conduction,wall} + R_{foul,o} + R_{convection,o}}$$



→ Inner convection resistance:-

$$R_{conv,i} = \frac{1}{h_i A_i} = \frac{1}{h_i(\pi D_i L)}$$

→ Wall conduction resistance (for cylinder):-

$$R_{cond,wall} = \frac{\ln(D_o/D_i)}{2\pi k_{wall} L}$$

→ Outer fouling resistance (based on outer area):-

$$R_{foul,o} = \frac{R_f}{A_o} = \frac{R_f}{\pi D_o L}$$

→ Outer Convection resistance:-

$$R_{conv,o} = \frac{1}{h_o A_o} = \frac{1}{h_o(\pi D_o L)}$$

The overall heat transfer coefficient $= U_o$

So, $R_{total} = \frac{1}{U_o A_o}$

So,

$$\frac{1}{U_o} = A_o\left(R_{conv,i} + R_{cond,wall} + R_{foul,o} + R_{conv,o}\right)$$

$$= \left(\frac{A_o}{h_i A_i}\right) + A_o\left(\frac{\ln(D_o/D_i)}{2\pi k_{wall} L}\right) + A_o\left(\frac{R_f}{A_o}\right) + \left(\frac{A_o}{h_o A_o}\right)$$

$$\therefore \boxed{\frac{1}{U_o} = \frac{D_o}{h_i D_i} + \frac{D_o \ln(D_o/D_i)}{2k_{wall}} + R_f + \frac{1}{h_o}}$$

Assumptions :-

i) Steady state condition :- Temperatures at any point in the system don't change with time. The rate of heat transfer (Q) is constant through every layer.

ii) One-dimensional radial conduction :- heat is assumed to flow only in the radial direction. Axial heat conduction (along the length) is considered negligible.

iii) Constant material properties :- The thermal conductivity of the wall (k_{wall}) is assumed to be constant and independent of temperature.

iv) Uniform convection coefficients :- The convective heat transfer coefficients (h_i and h_o) are assumed to be constant over the entire surface area of the tube.

v) No internal heat generation :- no heat generation due to electrical heating or chemical reactions.

vi) Adiabatic condition :- heat lost by the hot fluid is gained by the cold fluid, no heat losses to the external surroundings.

Part B :- Conceptual questions

$$\frac{1}{U_o} = R'_{total,o} = \underbrace{\frac{D_o}{h_i D_i}}_{\text{Inner convection}} + \underbrace{\frac{D_o \ln(D_o/D_i)}{2k_{wall}}}_{\text{Wall conduction}} + \underbrace{R_f}_{\text{Outer fouling}} + \underbrace{\frac{1}{h_o}}_{\text{Outer convection}}$$

$D_i = 0.020\,m$
$D_o = 0.040\,m$
$\frac{D_o}{D_i} = 2$

i) $R'_{conv,i} = \frac{D_o}{h_i D_i} = \frac{2}{h_i}$

$R'_{conv,o} = \frac{1}{h_o}$

for any scenario, where $h_i = h_o$, the inner resistance term will be twice as large as the outer resistance term.

Since, larger resistance means larger impact on U_o, h_i is the more performance limiting factor.

if $D_o >> D_i$ (i.e., D_o and D_i are very different), then h_i becomes even more dominant!

ii) Conduction resistance term, $R'_{cond,wall} = \frac{D_o \ln(D_o/D_i)}{2k_{wall}}$

So, decreasing the thermal conductivity (k) will increase the wall conduction resistance term. This leads to significant decrease in overall heat transfer coefficient term (U_o).

For a fixed duty, the required heat transfer area is inversely proportional to U_o ($A \propto 1/U_o$). So, switching from Copper to Stainless steel would require a much larger heat exchanger area for a given duty.

iii) Dominant resistance is the largest term in the sum of $1/U_o$ (Here, $D_i = 0.02\,m$, $D_o = 0.04\,m$, $\frac{D_o \ln(D_o/D_i)}{2} \approx 0.01386$)

(a) Inner convection dominates ($\frac{D_o}{h_i D_i}$ largest):-

Numerically,

$h_i = 50 W/m^2K$; $h_o = 1000 W/m^2K$; $k_{wall} = 385 W/mK$

$R_f = 0$; Here, h_i very small, h_o, R_f, k_{wall} are good/low.

Dominant term $= R'_{conv,i} = \frac{2}{50} = 0.04 m^2K/W$

(Total $= 0.041 m^2K/W$)

→ Performance bottlenecked by the inner fluid side. To improve heat exchanger, design effort should focus only on increasing h_i.

(b) Wall conduction dominates ($\frac{D_o \ln(D_o/D_i)}{2k_{wall}}$ largest):-

k_{wall} very small, h_i, h_o, R_f are good/low.

Numerically,

$h_i = 1500 W/m^2K$; $h_o = 1500 W/m^2K$; $k_{wall} = 16 W/mK$;

$R_f = 0$;

Dominant term $= R'_{cond,wall} = \frac{0.01386}{16} \approx 0.000866 m^2K/W$

(Total $\approx 0.002866 K/W$)

→ Changing the wall material to one with a higher k, will improve U_o.

(c) Fouling dominates (R_f largest):-

R_f is very large, h_i, h_o, k_{wall} are good/low.

Numerically,

$h_i = 2000 W/m^2K$; $h_o = 1500 W/m^2K$; $k_{wall} = 385 W/mK$;

$R_f = 0.005 m^2K/W$.

Dominant term $= R'_{foul} = 0.005 m^2K/W$ (Total $\approx 0.0067 m^2K/W$)

→ The exchanger is cleanable. Performance is limited by scale buildup. Maintenance focus should be on cleaning the outer surface to restore U_o.

iv) (a) U_o is most sensitive to small increase in R_f when exchanger is clean (i.e., R_f is close to zero).

→ U_o v/s R_f plot ($U_o \propto 1/(C+R_f)$) is steepest near the origin.

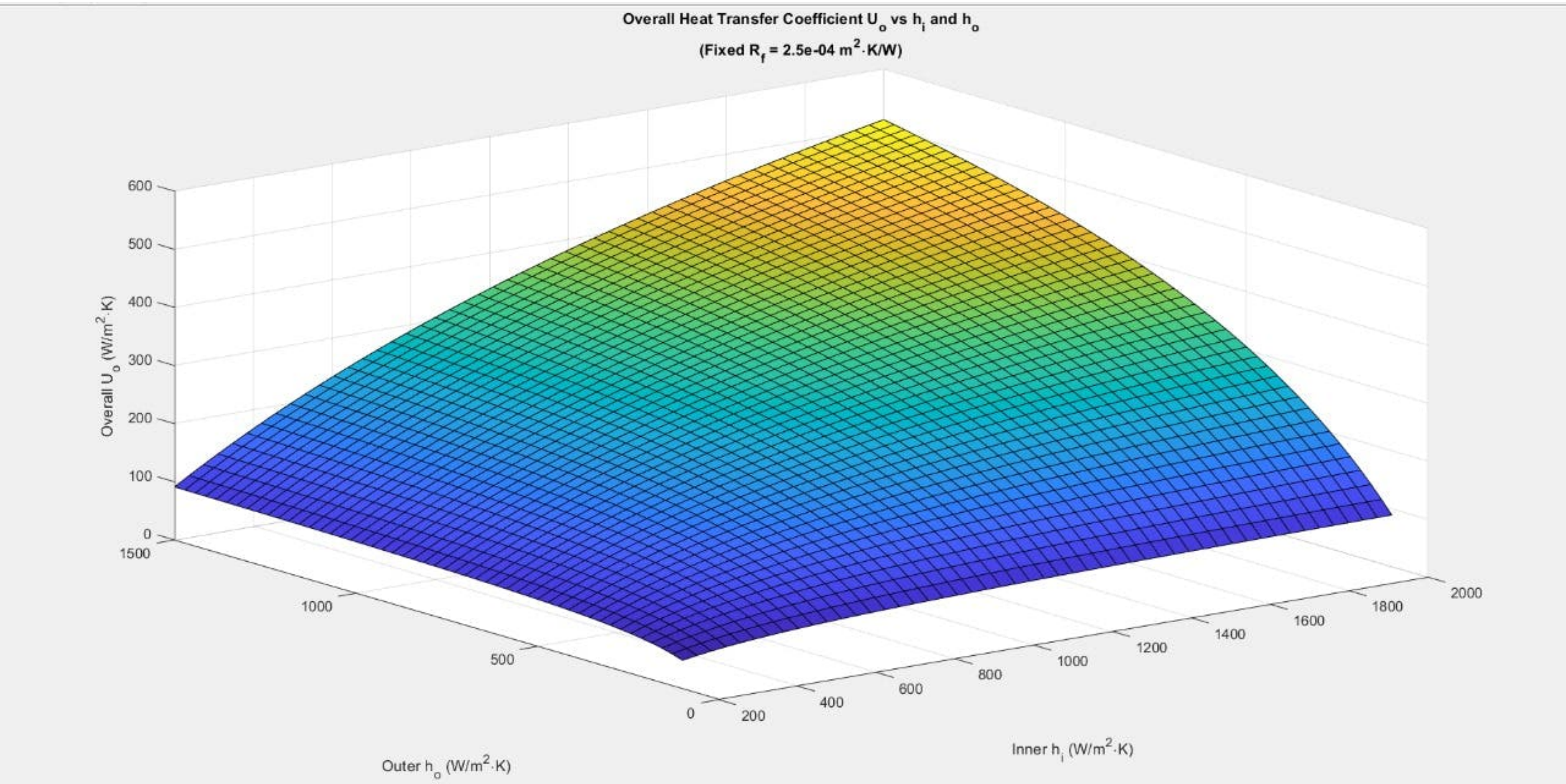
→ When R_f → small, total resistance is dominated by C. Adding a small R_f means a large fractional increase in R_{total}, causing a sharp initial drop in U_o.

→ as R_f increases, the curve flattens, i.e., U_o becomes less sensitive to higher values of R_f.

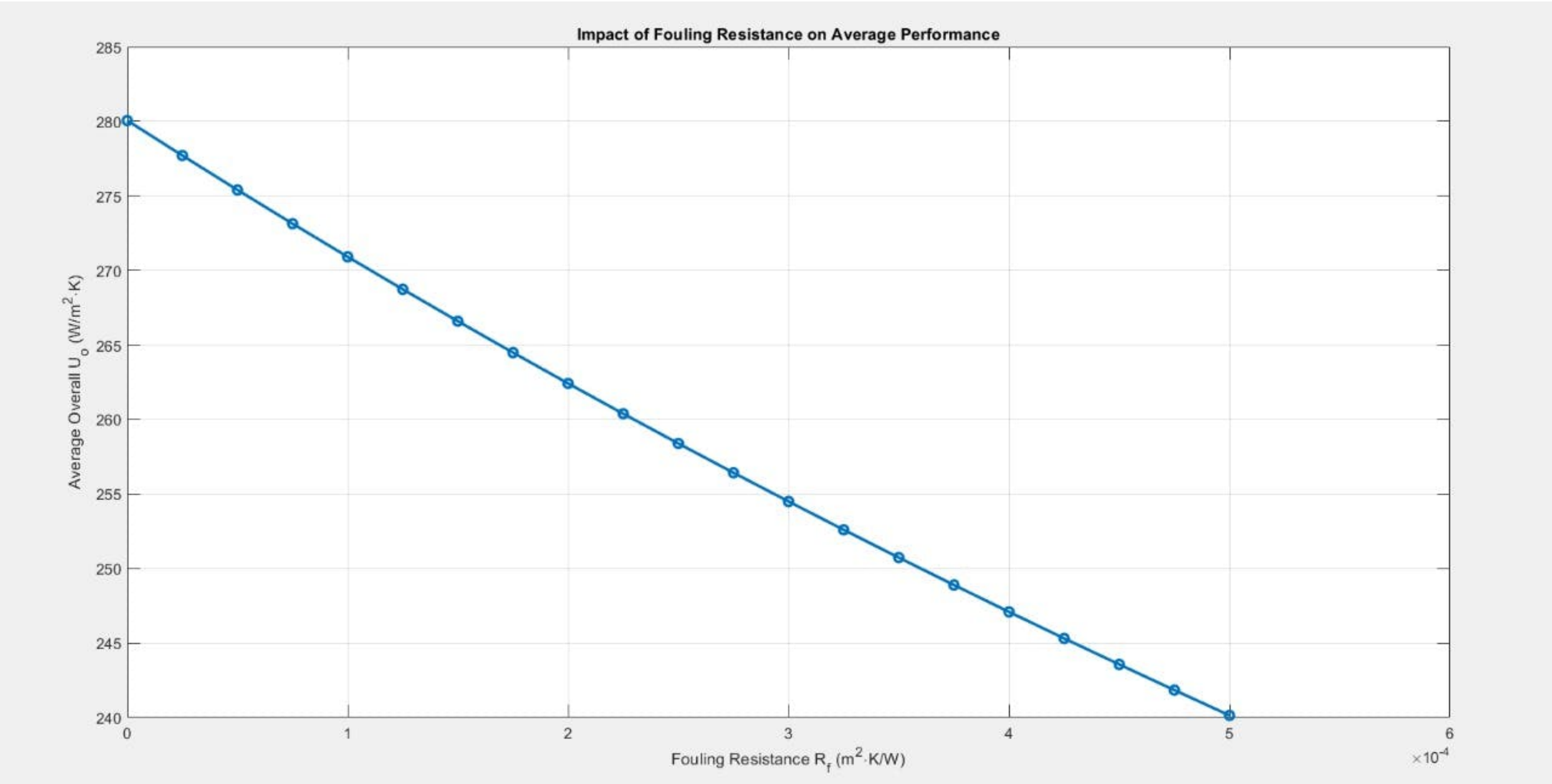
(b) Cleaning should be considered when overall heat transfer coefficient value U_o drops by 20% to 25% with respect to initial clean value (U_{clean} at $R_f = 0$).

Qualitative threshold (using the range $[0, 5\times10^{-4}]$):-

approximately around $R_f \simeq 3\times10^{-4}\ m^2K/W$. Beyond this point U_o loss is substantial.



Overall Heat Transfer Coefficient U_o vs h_i and h_o
(Fixed R_f = 2.5e-04 $m^2 \cdot K/W$)
Overall U_o ($W/m^2 \cdot K$)
Outer h_o ($W/m^2 \cdot K$)
Inner h_i ($W/m^2 \cdot K$)
0
100
200
300
400
500
600
1500
1000
500
0
200
400
600
800
1000
1200
1400
1600
1800
2000


Impact of Fouling Resistance on Average Performance
Average Overall U_o (W/m²·K)
Fouling Resistance R_f (m²·K/W)
×10⁻⁴
285
280
275
270
265
260
255
250
245
240
0
1
2
3
4
5
6